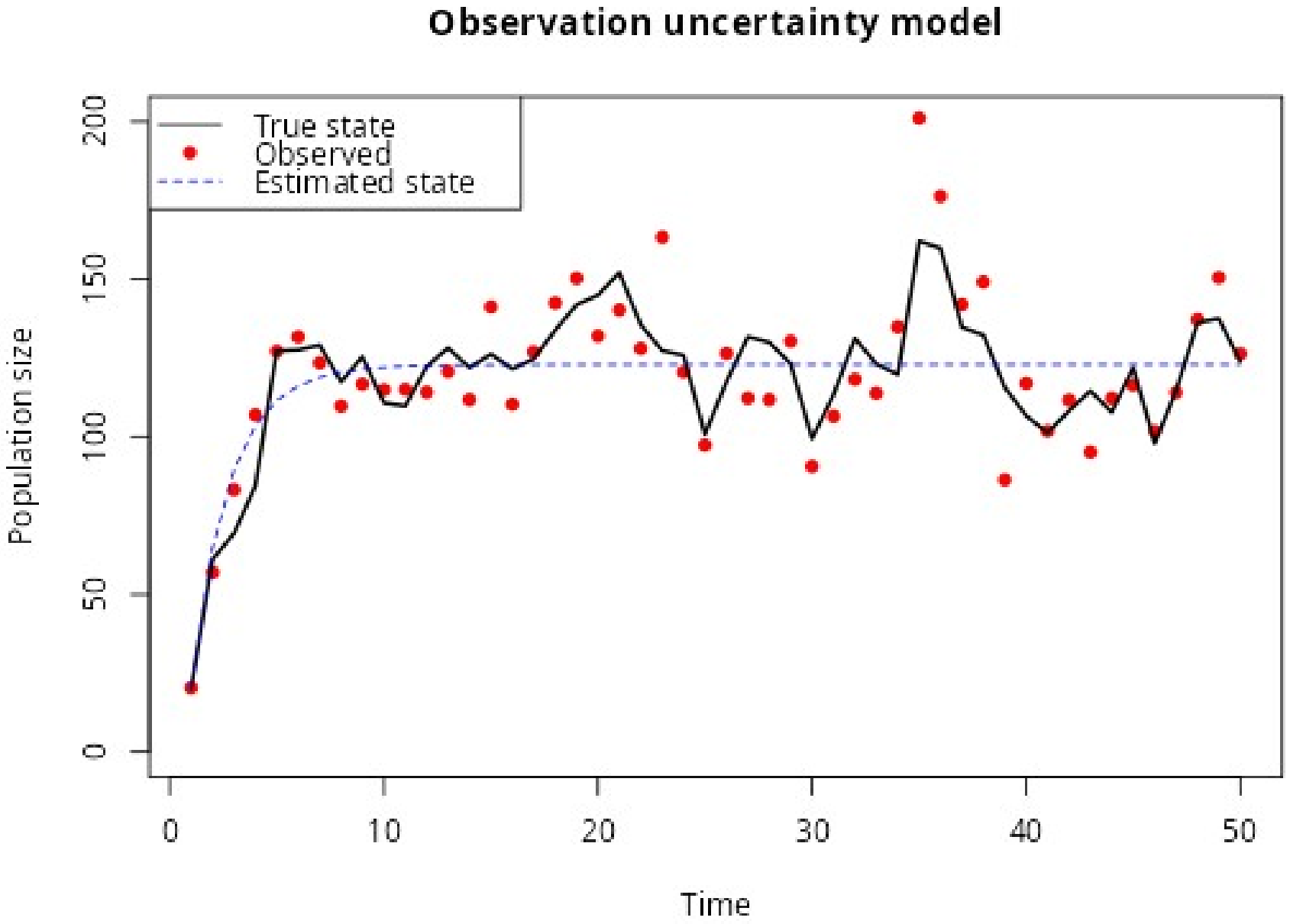
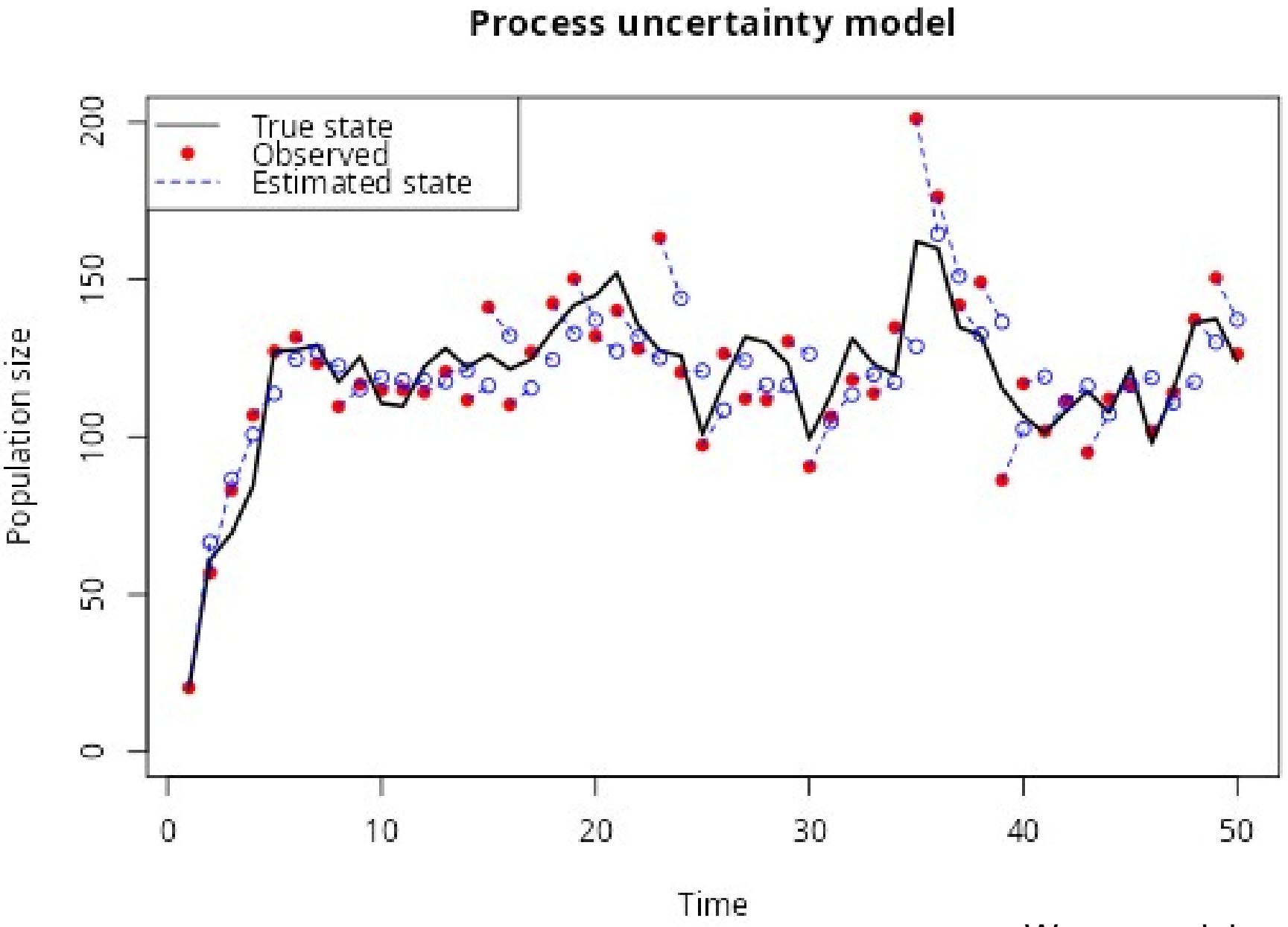
Today

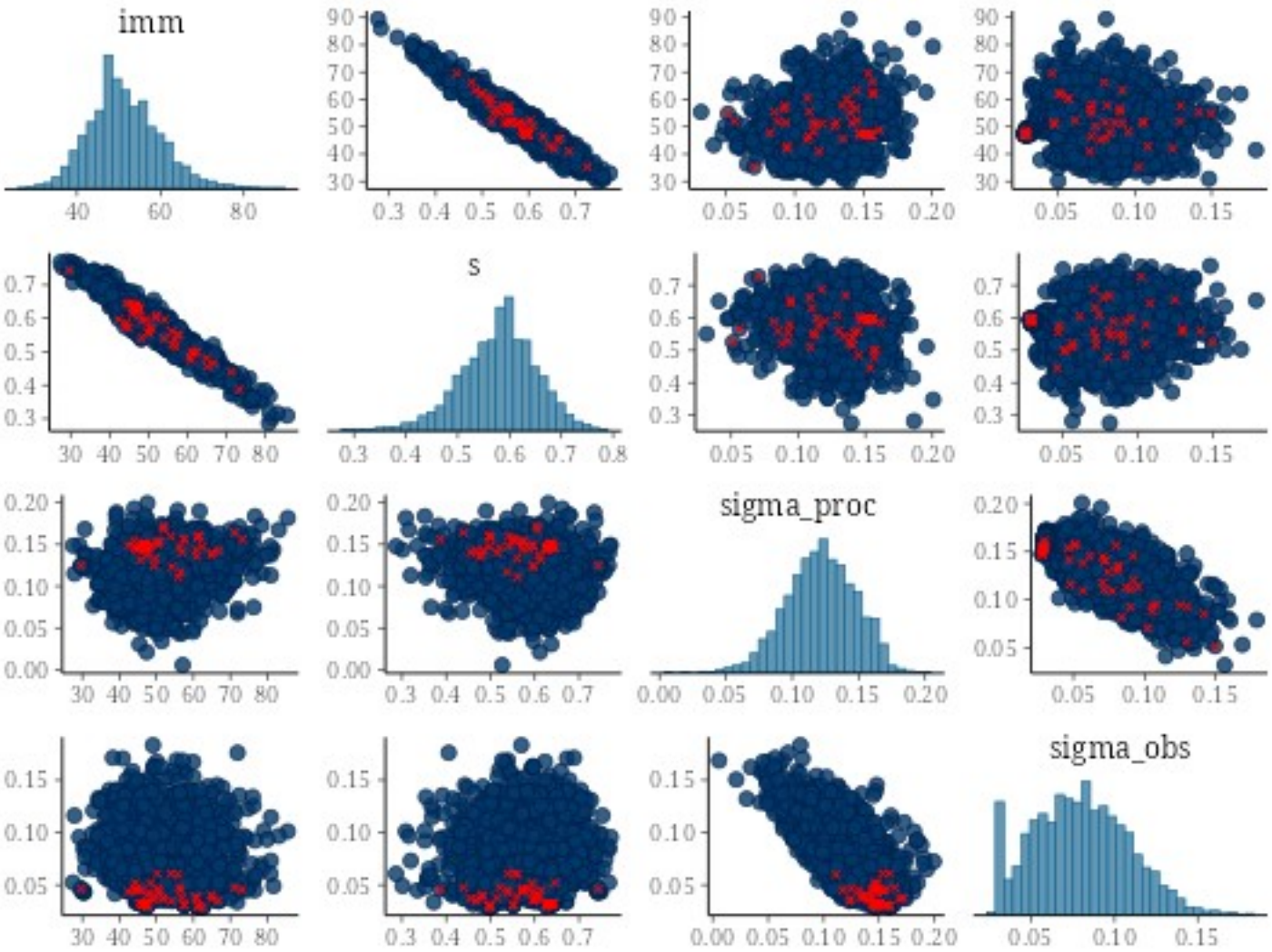
- State space models
- Code and training in Stan


Observation uncertainty model
True state
Observed
Estimated state
Population size
0
50
100
150
200
0
10
20
30
40
50
Time

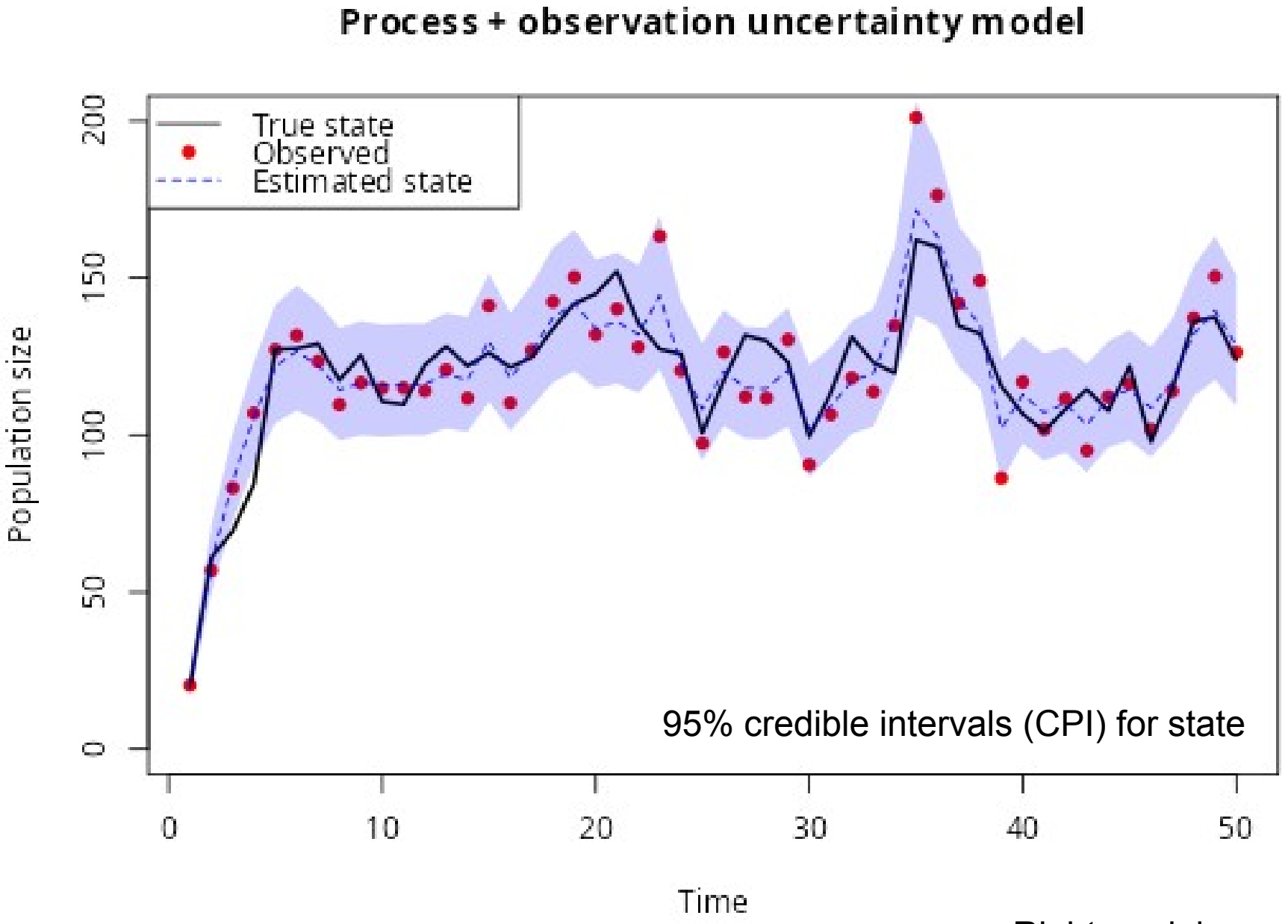
Wrong model


Process uncertainty model
True state
Observed
Estimated state
Population size
0
50
100
150
200
0
10
20
30
40
50
Time

Wrong model



Early fit of state space model: posterior sampling has issues


Process + observation uncertainty model
True state
Observed
Estimated state
Population size
0
50
100
150
200
0
10
20
30
40
50
Time
95% credible intervals (CPI) for state

Right model